MAXOPP - M18SL7 BOPP
Films

Max Speciality Films Ltd

Metallised Non Heat Sealable BOPP Film Inside Metallised Ultra High Oxygen Barrier Grade
for Adhesive & Extrusion Sandwich Lamination

PRODUCT SPECIFICATION SHEET (QA)

Properties	Position	M18SL7 Nominal Values*	Tolerance Limit	Unit	Method
• GENERAL					
Nominal Thickness	-	18	±5 %	µ	MAX
Grammage	-	16.38	±5 %	g/m^2	MAX
• OPTICAL					
Optical Density	-	2.9	≥2.76	-	MAX
• SURFACE					
Metal Adhesion	-	100	-	%	MAX
Metal Bond strength		700	≥ 500	Gms/25mm	AIMCAL
COF (Film/metal)		0.2	0.15-0.30	-	ASTM D 1894
COF (Film/Film)		0.45	0.35-0.60	-	ASTM D 1894
• MECHANICAL					
Tensile Strength	- MD	1200	≥1100	kg/cm^2	ASTM D 882
(at break)	- TD	2500	≥2200		
Elongation	- MD	190	≤230	%	ASTM D 882
(at break)	- TD	60	≤70		
Elastic Modulus	- MD	17000	≥16000	kg/cm^2	ASTM D 882
	- TD	28000	≥25000		
• THERMAL					
Linear Shrinkage	- MD	4	≤ 5	%	ASTM D 1204
	- TD	2	≤ 3		
• BARRIER					
WVTR 38° C (90% Rh)	-	0.10	≤ 0.13	$g/m^2/day$	ASTM F 1249
OXTR 23° C (0 % Rh)	-	0.10	≤ 0.13	$cc/m^2/day$	ASTM D 3985
• Defects					
Gels /Fish Eyes		> 0.80 mm-0 0.50-0.80 mm - 7 0.2-0.50 mm-10	> 0.80 mm-05 0.50-0.80 mm - 10 0.2-0.50 mm-15	No's per meter²	MAX
Sticky Marks		Sticky Marks shall Observe (No Impact on Barrier)	> 30 mm-00 no. 20-30 mm -3 no. 10-20mm0-3 no. < 10 mm-6 no	No's meter²	MAX
Tram line		Tram Lines shall Observe (No Impact on Barrier)	Can Observe at 1-2 places across width	No's	MAX
Pin Holes		3.0	5	Per A-4	MAX
Scratches (Base Film)		10	20	No's per A-4	MAX
		20	< 30	mm (Size)	MAX
Blocking**		Level-00	Level-01	Rating	MAX
Ripples**		Level-01	Level-03	Rating	MAX
Sagging**		Level-02	Level-02	Rating	MAX
• Remarks					
Width Tolerence		- 0 & + 2mm			
Recmended Extrusion GSM		≤ 12 Gm/sq.m			

Joints	Single Joint	Percentage	Order Qty (in Tons)	Double Joint	Percentage	Order Qty (in Tons)	Remarks
		20% 15% 10%	≤ 1.0 1.0-5.0 ≥ 5.0		5% 5% 5%	≤ 1.0 ≤ 5.0 ≥ 5.0	TYPE OF JOINT - LAP JOINT

Corona Treatment (Only for Non-metal side surface)	Minimum 38 dynes	Minimum 36 dynes	-	Remarks
	Three months	Six months		

Test ceritificate	Test ceritificate Shall Be Provided with Each Consignment
Treatment side	Shall be indicated in each Roll With Sticker
Manufacturing Date	Shall be indicated in each roll Slip

Shelf Life : Film characteristics are maintained (Except dyne level on Metallised surface) for six months from the invoice date.

We ensure dynes value more than 38 dynes/cm at the time of material leaving our factory. For better results, we strongly recommend online corona treatment in metalized films during lamination. Treatment level decay with time is a natural phenomenon which depends on ambient conditions (Recommended storage conditions: Temperature < 30 ℃ & Humidity 55% (Maximum) in original packed condition).

* The figures and above properties refer to average values of laboratory test on samples of our standard production with specified lab equipements as per ASTM/MAX/AIMCAL test methods only. Customers should verify the suitability of the film for its specific end use.
Keep rolls in hanging conditions only; Material prone to blocking. (Please refer MSFL product handling guidelines during handling of this product).
** Marked Defects level increses with ageing of products.
Max and Max Flame logo, registered trademarks of Max India Limited, are being used by Max Speciality Films Limited under license Agreements.

Approved By

Dated